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0.2 | Bearing Shape Factor

The stiffness of rubber in compression, when the loaded surfaces are prevented from slipping by bonding or by mechanical location, depends upon the shape factor S defined as the ratio of one loaded area to the total force-free area (see Fig. 1). [0.2.1]

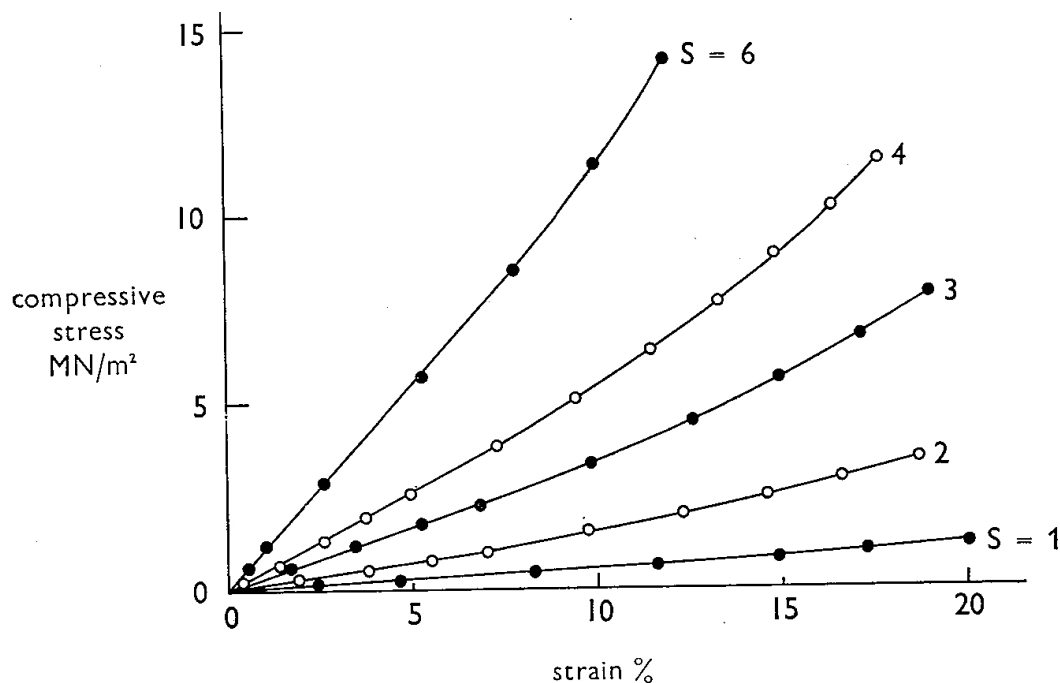


Fig. 1 - Shape Factor

Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

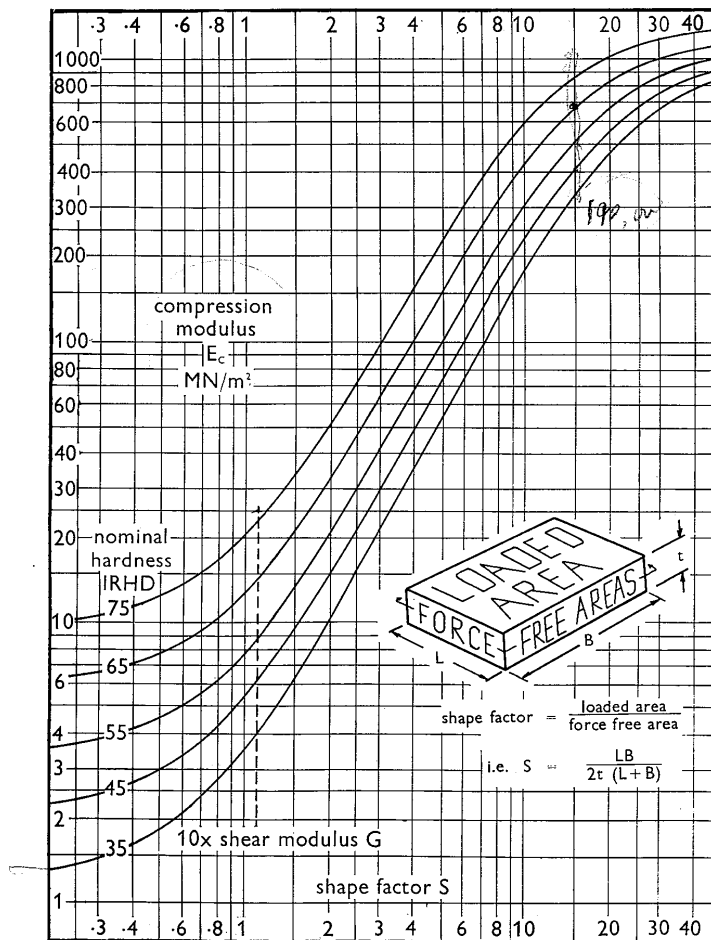


Fig. 28.—Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above).

* Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the E_∞ value in Table 3 should be doubled.

Fig. 2 - Compression Modulus

Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above). Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred then the E_∞ value in Table 3 should be doubled.

Table 1: IRHD hardness scale (MN/m²) (stored: t002-1.csv)

Hardness	Young's mod E0	Shear mod G	k	Bulk mod
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in²) (stored: t002-2.csv)

Hardness	Young's mod E0	Shear mod G	k	Bulk mod
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Notes

- k is used in the calculation of compression characteristics.
- The majority of springs are in the hardness range 40-60 IRHD.
- Average design limits: 15% compression, 50% shear.
- Theoretically, with a Poisson's ratio of 1/2, E0 should equal 3G. This is so for soft gum rubbers, but for harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase E0 to about 4G.
- Based on experiments on natural rubber spring vulcanizates containing (above 48 IRHD) SRF black as filler. Note that hardness is subject to an uncertainty of about +/-2 degrees.



0.2 - 2 | Block Compression Stiffness

Table 3: Shape Factor Parameters

variable	value	[value]	description
t ₁	0.25	0.25	thickness
L ₁	24	24	length
B ₁	12	12	breadth

For a rectangular section (length = L and breadth = B):

Eq. 1: shape factor

$$S_1 = \frac{B_1 \cdot L_1}{2 \cdot t_1 \cdot (B_1 + L_1)}$$

$S_1 = 16.0 \quad [S_1] = 16.0 \quad | \text{ shape factor}$

B_1	t_1	L_1
12.0	0.25	24.0
breadth	thickness	length

For a block of square section (i.e. L = B) or circular section (diameter = L):

Eq. 2: shape factor

$$S_2 = \frac{L_1}{4 \cdot t_1}$$

$S_2 = 24.0 \quad [S_2] = 24.0 \quad | \text{ shape factor}$

t_1	L_1
0.25	24.0
thickness	length

The compression modulus E_c depends upon the shape factor S (for derivation see Section 5).

Table 4: E_c parameters



variable	value	[value]	description
E0	100.00 p_si	0.69 MPA	compresssion modulus
k1	0.7	0.7	adjustment factor
G1	0.50 k_si	3.45 MPA	shear modulus

Eq. 3: compression modulus

$$E_c = E_0 \cdot \left(2 \cdot S_1^2 \cdot k_1 + 1 \right)$$

$E_c = 35.94 \text{ k_si}$
 $[E_c] = 247.80 \text{ MPA}$
| compression modulus

S_1	E_0	k_1
16.0	100.00 p_si	0.7
shape factor	compresssion modulus	adjustment factor

When $S > 3$ it may be more convenient to use:

Eq. 4: compression stiffness - modified

$$E_{lc} = 5 \cdot G_1 \cdot S_1^2$$

$E_{lc} = 640.00 \text{ k_si}$
 $[E_{lc}] = 4412.64 \text{ MPA}$
| compression stiffness - modified

G_1	S_1
0.50 k_si	16.0
shear modulus	shape factor

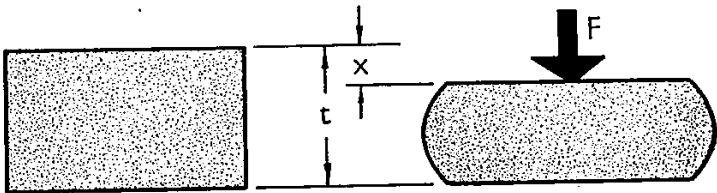


Fig. 29.—Rubber block in compression.

Fig. 3 - Compression Stiffness K_c



The diagram shows a rectangular rubber block of thickness t , deflecting by x under a compressive load F applied to a cylindrical rubber sample.

The compression stiffness K_c is given by:

Eq.5:

$$K_c = \frac{F}{x}$$

Table 5: Stiffness parameters

variable	value	[value]	description
A1	20.00 in ²	12903.20 mm ²	cross-sectional area
t1	0.25 inch	6.35 mm	thickness
x1	1.00 inch	25.40 mm	deflection

Eq. 6: compression stiffness

$$K_c = \frac{A_1 \cdot E_c}{t_1}$$

$$K_c = 2875.20 \text{ k_in} \quad [K_c] = 5035.25 \text{ kN_cm} \quad | \text{ compression stiffness}$$

t_1	E_c	A_1
0.25 inch	35940.00 p_si	20.00 in ²
thickness	compression modulus	cross-sectional area

If $S > \mu/2$, where S is the shape factor and μ the coefficient of friction, slip may occur. In such cases, the rubber should be bonded to the rigid

The load (F)—deflexion (x) curve of rubber in compression is non-linear. With no slip it has the approximate form

Eq.7:

$$F = A \cdot E_c \cdot e \cdot (e + 1)$$

where e , the compressive strain, equals x/t . The non-linearity is usually ignored for strains up to about 10%.



There is as yet no method of calculating that a block will be stable but experience has shown that provided the thickness is less than one-quarter of the least plan dimension there should be no instability.

0.2 - 3 | Strip Compression Stiffness

When a long strip of rubber is compressed the strain in the direction of its length will be negligible.

Shape factor

Eq.8:

$$SF = \frac{b}{2 \cdot t}$$

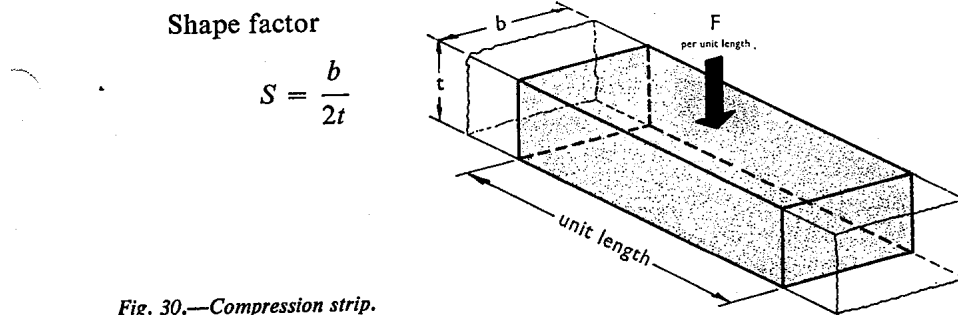


Fig. 4 - Compression Strip

Diagram shows a long rectangular strip of rubber of width b , thickness t , and unit length, compressed by a load F applied per unit length.

Compression modulus E_c

Eq.9:

$$E_c = \frac{4 \cdot E_0 \cdot \left(SF^2 \cdot k + 1 \right)}{3}$$

The compression stiffness per unit length, K_c , is given by

Eq.10:

$$K_c = \frac{F}{x}$$

Eq.11:

$$K_c = \frac{E_c \cdot b}{t}$$

Eq.12:

$$K_c = \frac{4 \cdot E_0 \cdot b \cdot \left(SF^2 \cdot k + 1 \right)}{3 \cdot t}$$

F = load per unit length

E_c = compression modulus (corrected, if necessary, for the effect of bulk compression)E₀ = Young's modulus (from Table 3)

b = width of strip

t = thickness of strip

x = deflexion

k = a numerical factor (from Table 3)

SF = shape factor

The load per unit length (F)—deflexion (x) curve for a compressed strip is non-linear. It has the approximate form

Eq.13:

$$F = E_c \cdot b \cdot e \cdot \left(\frac{3 \cdot e}{2} + 1 \right)$$

where e, the compressive strain, equals x/t. As in the case of blocks, the non-linearity is usually ignored for strains up to about 10%.



0.2 - 4 | Derivation of Compression Characteristics

Approximate load-deformation relations are derived below for small compressions of rubber blocks, between rigid plates to which they adhere or are bonded. The total displacements are considered to arise from the superposition of simple displacements, namely, (1) the pure homogeneous deformation defined by the displacement of one rigid bonding plate towards the other, and (2) the subsequent displacements necessary to cause points in the planes of the bonded surfaces to be restored to their original positions in these planes. [0.2.2]

Two extreme cases are considered, infinitely long rectangular blocks, the infinite dimension being at right angles to the direction of compression, and circular disks, loaded axially.

Rectangular Blocks of Infinite Length **_bold**

The pure homogeneous deformation consists of a compressive strain e in the vertical direction, zero strain in the length direction, and an expansion in the width direction given, since the rubber is virtually incompressible in volume, by e also. The rubber is placed in a state of pure shear and the force F_1 which has to be applied to the bonded surfaces to maintain such a deformation is given by

Eq.14:

$$F_1 = \frac{4 \cdot E_0 \cdot b \cdot e}{3}$$

per unit length, where E_0 is the Young's modulus of the rubber and b is the width of the block.

The second displacement system is that causing points in the planes of the bonded surfaces to be restored to their original positions in these planes. In order to calculate the corresponding system of forces which has to be applied to the bonded surfaces, it is necessary to make the simplifying assumption that horizontal planes remain plane during the deformation, as seems probable when the width b is much greater than the thickness t . Vertical sections consequently take up parabolic forms, as represented diagrammatically in Fig. 41.

The maximum outward displacement of any originally vertical section is dictated by the requirement that the volume contained between it and the central plane in the undeformed state shall equal the volume contained between it and the central plane when it takes up the parabolic form and the bonded surfaces approach by an amount $\delta (= e \cdot t)$. [0.2.3]

For a plane at a distance x from the central vertical plane, considering unit length, we have

Eq.15:

$$t \cdot x = \frac{2 \cdot t \cdot u}{3} + x \cdot (-\delta + t)$$



where $2ut/3$ is the approximate volume contained between the parabolic front of maximum displacement u and the vertical plane at x , when the compression δ is small. Hence,

Eq.16:

$$u = \frac{3 \cdot \delta \cdot x}{2 \cdot t}$$

In an elementary section of width dx the small displacement u may be maintained by the action of an excess hydrostatic pressure dp acting on one curved face, where dp is given by the classical theory of elasticity in the form

Eq.17:

$$dp = dx \cdot \frac{-8 \cdot E_0 \cdot u}{3 \cdot t^2}$$

Since the curved surface at $x = b/2$ is force free, p may be obtained by integrating both sides of equation (2) between the limits $x = x$ and $x = b/2$, yielding

Eq.18:

$$p = \frac{2 \cdot E_0 \cdot \delta \cdot \left(\frac{b^2}{4} - x^2 \right)}{3 \cdot t}$$

The pressure p acts on the rigid bonded plates also. The corresponding force F_2 acting on each of these surfaces may be obtained by integrating such terms as $p \cdot dx$ between the limits $x = b/2$ and $x = -b/2$, to give

Eq.19:

$$F_2 = \frac{E_0 \cdot b \cdot \delta^3}{3 \cdot t}$$

The apparent value of Young's modulus for the rubber block is given by



Eq.20:

$$\square = F \cdot \frac{1}{b} \cdot \frac{1}{e}$$

where F is the total force which has to be applied to the bonded surfaces per unit length to maintain the deformation, and is given by $F_1 + F_2$ where F_1 and F_2 are given by the equations. Hence,

Eq.21:

$$\square = \frac{E_0 \cdot b^2}{2 \cdot t} + \frac{4 \cdot E_0}{3}$$

where the shape factor S , the ratio of one loaded surface to the force-free surface, is equal to $b/(2 \cdot t)$

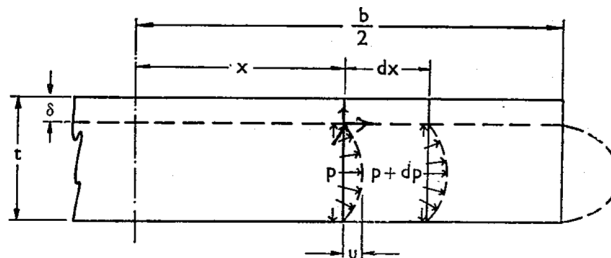


Fig. 5 - Initial and Deformed States

Initial and deformed states for an infinitely long block subjected to a small compression.

Circular Disks

The apparent Young's modulus may be derived similarly for a circular disk. The force F_1 which must be applied to maintain the first deformation, a simple compression of amount e , is given by [0.2.4]

Eq.22:

$$F_1 = E_0 \cdot a^2 \cdot e \cdot \pi$$

where a is the radius of the disk.



The force F_2 which has to be applied to maintain the second displacement system may be calculated by a similar procedure to that used in the previous example. Initially, vertical cylindrical surfaces are assumed to take up parabolic forms such that the volume contained within them is unchanged. When the compressive displacement δ is small, and the radius r is much greater than the maximum radial displacement u , then

Eq.23:

$$u = \frac{3 \cdot \delta \cdot r}{4 \cdot t}$$

where t is the thickness of the disk. The pressure p acting at a distance r from the centre is found to be

Eq.24:

$$p = \frac{E_0 \cdot \delta \cdot \left(\frac{a^2}{t^3} - r^2 \right)}{t^3}$$

and hence the force F_2 is obtained as

Eq.25:

$$F_2 = \frac{E_0 \cdot a^4 \cdot \delta \cdot \pi}{2 \cdot t^3}$$

An analogous relation has been derived by Dienes and Klemm for the rate of approach of two parallel circular plates separated by a viscous liquid.

The apparent Young's modulus E for a circular disc is obtained finally in the form

Eq.26:

$$E = E_0 \cdot \left(\frac{a^2}{2 \cdot t^2} + 1 \right)$$



where S , the ratio of one loaded surface to the force-free surface, is equal to $a/(2t)$.

As the thickness of the block is increased, the terms representing the contribution F_2 to the total force decrease, so that when the condition on which the calculation for F_2 is based – that the width or radius is much greater than the thickness – becomes inapplicable. The contribution F_2 to the total force becomes relatively small. The equations may therefore be expected to apply over a considerable range of thicknesses.

NOTE: In practice it is necessary to multiply the S^2 term by the factor k , given in Table 1. [0.2.5]

[0.2.1]

P.B. Lindley, Engineering Design with Natural Rubber, NR Technical Bulletin. Malaysian Rubber Producers' Research Association, Brickendonbury, U.K. pages 26-28, 1959.

[0.2.2]

P.B. Lindley, Engineering Design with Natural Rubber, NR Technical Bulletin. Malaysian Rubber Producers' Research Association, Brickendonbury, U.K. pages 45-46, 1959.

[0.2.3]

From Appendix I of The Compression of Bonded Rubber Blocks, by A. N. Gent and P. B. Lindley, Proceedings of the Institution of Mechanical Engineers, 1959, Vol. 173, No. 3, pages 111-122.

[0.2.4]

SOUTHWELL, R. V. 1941 'An Introduction to the Theory of Elasticity', p. 126 (Clarendon Press Oxford). ADKINS, J. E. 1954 Proc. Camb. phil. Soc., vol. 50, p. 334.

[0.2.5]

DIENES, G. J. and KLEMM, H. F. 1946 J. Appl. Phys., vol. 17, p. 458.

